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Create a vector, scalar, matrix and tensor with values of your choosing using `tf.constant()`. Find the shape, rank and size of the tensors you created in 1. Create two tensors containing random values between 0 and 1 with shape `[5, 300]`. Multiply the two tensors you created in 3 using matrix multiplication. Multiply the two tensors you created in 3 using dot product. Create a tensor with random values between 0 and 1 with shape `[224, 224, 3]`. Find the min and max values of the tensor you created in 6. Created a tensor with random values of shape `[1, 224, 224, 3]` then squeeze it to change the shape to `[224, 224, 3]`. Create a tensor with shape `[10]` using your own choice of values, then find the index which has the maximum value. One-hot encode the tensor you created in

```
import tensorflow as tf

# Create a scalar
scalar = tf.constant(7)
# Create a vector
vector = tf.constant([1, 2, 3])
# Create a matrix
matrix = tf.constant([[12, 23], [1, 4]])
# Create a tensor (e.g., a 3D tensor)
tensor = tf.constant([[[1, 2, 3], [4, 5, 6]], [[7, 8, 9], [10, 11, 12]]])

print("Scalar:", scalar)
print("Vector:", vector)
print("Matrix:", matrix)
print("Tensor:", tensor)
```

```
Scalar: tf.Tensor(7, shape=(), dtype=int32)
Vector: tf.Tensor([1 2 3], shape=(3,), dtype=int32)
Matrix: tf.Tensor(
[[12 23]
 [ 1  4]], shape=(2, 2), dtype=int32)
Tensor: tf.Tensor(
[[[ 1  2  3]
 [ 4  5  6]]

 [[ 7  8  9]
 [10 11 12]]], shape=(2, 2, 3), dtype=int32)
```

Find the shape, rank, and size of the tensors

```
print("\n--- Scalar Properties ---")
print("Scalar Shape:", scalar.shape)
print("Scalar Rank:", tf.rank(scalar))
print("Scalar Size:", tf.size(scalar))

print("\n--- Vector Properties ---")
print("Vector Shape:", vector.shape)
print("Vector Rank:", tf.rank(vector))
print("Vector Size:", tf.size(vector))

print("\n--- Matrix Properties ---")
print("Matrix Shape:", matrix.shape)
print("Matrix Rank:", tf.rank(matrix))
print("Matrix Size:", tf.size(matrix))

print("\n--- Tensor Properties ---")
print("Tensor Shape:", tensor.shape)
print("Tensor Rank:", tf.rank(tensor))
print("Tensor Size:", tf.size(tensor))
```

```
--- Scalar Properties ---
Scalar Shape: ()
Scalar Rank: tf.Tensor(0, shape=(), dtype=int32)
Scalar Size: tf.Tensor(1, shape=(), dtype=int32)

--- Vector Properties ---
Vector Shape: (3,)
Vector Rank: tf.Tensor(1, shape=(), dtype=int32)
Vector Size: tf.Tensor(3, shape=(), dtype=int32)

--- Matrix Properties ---
Matrix Shape: (2, 2)
Matrix Rank: tf.Tensor(2, shape=(), dtype=int32)
Matrix Size: tf.Tensor(4, shape=(), dtype=int32)
```

```

--- Tensor Properties ---
Tensor Shape: (2, 2, 3)
Tensor Rank: tf.Tensor(3, shape=(), dtype=int32)
Tensor Size: tf.Tensor(12, shape=(), dtype=int32)

```

```

b = tf.constant([[1,2],[1,3]])
print("Tensor b:", b)

```

```

import tensorflow as tf

# Create two tensors containing random values between 0 and 1 with shape [5, 300]
x = tf.random.uniform(shape=[5, 300], minval=0., maxval=1.)
y = tf.random.uniform(shape=[5, 300], minval=0., maxval=1.)

```

```
tf.matmul(x, tf.transpose(y))
```

```

<tf.Tensor: shape=(5, 5), dtype=float32, numpy=
array([[75.65553 , 76.854065, 73.781334, 79.10141 , 79.40784 ],
       [72.677826, 77.65373 , 74.44951 , 74.52515 , 79.92966 ],
       [69.5861 , 75.910484, 71.99158 , 74.630646, 75.002655],
       [75.26797 , 78.61142 , 77.594894, 80.0971 , 80.91611 ],
       [69.235344, 73.81564 , 74.44196 , 72.15712 , 75.70548 ]],
      dtype=float32)>

```

```
tf.tensordot(x, tf.transpose(y), axes=1)
```

```

<tf.Tensor: shape=(5, 5), dtype=float32, numpy=
array([[75.65553 , 76.854065, 73.781334, 79.10141 , 79.40784 ],
       [72.677826, 77.65373 , 74.44951 , 74.52515 , 79.92966 ],
       [69.5861 , 75.910484, 71.99158 , 74.630646, 75.002655],
       [75.26797 , 78.61142 , 77.594894, 80.0971 , 80.91611 ],
       [69.235344, 73.81564 , 74.44196 , 72.15712 , 75.70548 ]],
      dtype=float32)>

```

```

import tensorflow as tf

# Create two tensors containing random values between 0 and 1 with shape [5, 300]
x = tf.random.uniform(shape=[224, 224, 3], minval=0., maxval=1.)
y = tf.random.uniform(shape=[224, 224, 3], minval=0., maxval=1.)

```

```
tf.reduce_min(x)
```

```
<tf.Tensor: shape=(), dtype=float32, numpy=1.7881393432617188e-06>
```

```
tf.reduce_max(y)
```

```
<tf.Tensor: shape=(), dtype=float32, numpy=0.9999997615814209>
```

```

import tensorflow as tf

# Create two tensors containing random values between 0 and 1 with shape [5, 300]
x = tf.random.uniform(shape=[1, 224, 224, 3], minval=0., maxval=1.)

```

```
tf.squeeze(x)
```

```

<tf.Tensor: shape=(224, 224, 3), dtype=float32, numpy=
array([[ [0.9743917 , 0.87352157, 0.30628228],
        [0.56202364, 0.6416595 , 0.86358714],
        [0.29258633, 0.22974932, 0.8205981 ],
        ...,
        [0.56623256, 0.04326057, 0.5969602 ],
        [0.43624222, 0.61775887, 0.56348217],
        [0.58343446, 0.25700104, 0.09418678]],

       [[0.42979527, 0.2811172 , 0.7836844 ],
        [0.7510896 , 0.7666873 , 0.34250832],
        [0.1521014 , 0.5020666 , 0.94051826],
        ...,
        [0.95977986, 0.93418455, 0.05378366],
        [0.47357976, 0.64933264, 0.13636076],
        [0.52803063, 0.07964444, 0.42493546]],

       [[0.17618692, 0.72865784, 0.31007576],
        [0.07402039, 0.57750297, 0.52093565],

```

```
[0.23578155, 0.35632682, 0.6105741 ],
...,
[0.3767115 , 0.9869664 , 0.93362284],
[0.34419072, 0.08158123, 0.60782254],
[0.8872404 , 0.11203265, 0.7525561 ]],

...,

[[0.9521836 , 0.7947203 , 0.09846401],
 [0.53137887, 0.6626345 , 0.05175161],
 [0.58131313, 0.78551626, 0.6721953 ],
 ...,
 [0.94556856, 0.9133154 , 0.35858953],
 [0.846552 , 0.4295255 , 0.29603446],
 [0.4788345 , 0.7532873 , 0.21702862]],

[[0.7181237 , 0.303344 , 0.727285 ],
 [0.9836867 , 0.23520672, 0.35201252],
 [0.55676746, 0.55726385, 0.40034664],
 ...,
 [0.07805789, 0.62478185, 0.19388783],
 [0.93229616, 0.9207821 , 0.93305147],
 [0.14614463, 0.3545692 , 0.8532442 ]],

[[0.41941166, 0.27130783, 0.68745744],
 [0.24304593, 0.3463782 , 0.11814821],
 [0.8811859 , 0.63591695, 0.7738223 ],
 ...,
 [0.24039507, 0.52121663, 0.42366076],
 [0.47660863, 0.9317298 , 0.28196025],
 [0.17327905, 0.37476373, 0.8511739 ]]], dtype=float32)>
```

```
g=tf.constant(np.random.randint(1,10,10))
print(g)
```

```
tf.Tensor([2 9 6 5 6 7 9 9 1 9], shape=(10,), dtype=int64)
```

```
tf.argmax(g)
```

```
<tf.Tensor: shape=(), dtype=int64, numpy=1>
```

```
tf.one_hot(g,depth=4)
```

```
<tf.Tensor: shape=(10, 4), dtype=float32, numpy=
array([[0., 0., 1., 0.],
       [0., 0., 0., 0.],
       [0., 0., 0., 0.],
       [0., 0., 0., 0.],
       [0., 0., 0., 0.],
       [0., 0., 0., 0.],
       [0., 0., 0., 0.],
       [0., 0., 0., 0.],
       [0., 1., 0., 0.],
       [0., 0., 0., 0.]], dtype=float32)>
```